

Organizational memory influences new product success

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Abstract

Prior research on the impact of organizational memory on new product success has divergent perspectives. Such inconsistency has accrued mainly from not considering memory's multifaceted aspects, which interact with specific project characteristics. This paper tries to sort out this paradox by proposing that project innovativeness moderates the relationship among variables of organizational memory and new product success. An empirical study of 169 NPD projects of Korean manufacturing firms finds that memory sharing and the use of external information and formal procedures enhance new product success, whereas organizational memory has no effect. Project innovativeness is found to moderate memory's effect, despite some tradeoffs. As innovativeness increases, the effect of organizational memory and use of external information become stronger whereas the effect of memory sharing and use of formal procedures weaken. This suggests that firms must activate organizational memory more carefully according to project characteristics and the memory level to maximize its positive effects.

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1. Introduction

New product development (NPD) is one of the key driving forces for attaining competitive advantage and sustained growth of the firm. Because of this, many scholars and practitioners struggle to find the factors that lead to NPD success. The importance of NPD notwithstanding, it seems there are no explicit practical guidelines for successful product development. Part of the answer lies in the fact that the NPD process includes a complex interaction with organizational factors and external environmental factors. That is, the process itself entails great uncertainty (Kanter, 1988; Van de Ven, 1986).

In this respect, many organizational learning theorists view NPD as a learning process because its problem-solving aspects turn individual team members' knowledge into collective team knowledge (Argyris and Schön, 1978; Imai et al., 1985; Sinkula, 1994). Some scholars contend that a learning capacity that surpasses that of competitors is the only source of a firm's

competitive advantage (De Geus, 1988; Dickson, 1992; Slater and Narver, 1995). This leads to the recognition that organizational learning and information processing abilities are the crucial sources of NPD success (Leonard-Barton, 1992; Lynn et al., 2000; Madhavan and Grover, 1998). Huber (1991) argues that organizational learning processes follow four distinguishable phases: knowledge acquisition, information distribution, information interpretation, and organizational memory. Researchers focus mainly on the first three stages, whereas relatively little attention is paid to organizational memory as the storage of knowledge that results from learning.

Scholarly research results also differ on the impact of organizational memory on performance. Researchers who champion the positive effects insist that organizational memory should enhance performance because a firm's long-standing experience promotes efficiency (Cyert and March, 1963; Duncan and Weiss, 1979). Other researchers point to memory's deleterious effects, observing that change from prior patterns becomes more difficult when memory in a particular domain increases (Dougherty, 1992; Leonard-Barton, 1992). They indicate that the effect is not straightforward, but has a more complex and contingent role in product development. Some also argue that the way memory is manifested and shared in organizations is more

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